Soil Moisture Report

Green Farm · Tub 1 · Wed 4 Feb to Tue 10 Feb
Generated 10 Feb 2026, 09:49 SAST

This week, do this

1 **Water Tub 1 today (Tue 10 Feb)**: about 39 minutes (about 58 L).
High confidence
Why: Your soil has dried to the point where tomato starts to struggle, and no rain is expected today. (note 1)

2 **Reset Tub 1's watering level**: it waits until the soil is drier than this crop handles well. Tap Reset to recommended in the app.
High confidence
Why: The watering level saved for this zone waits until the soil is drier than tomato handles well. (note 2)

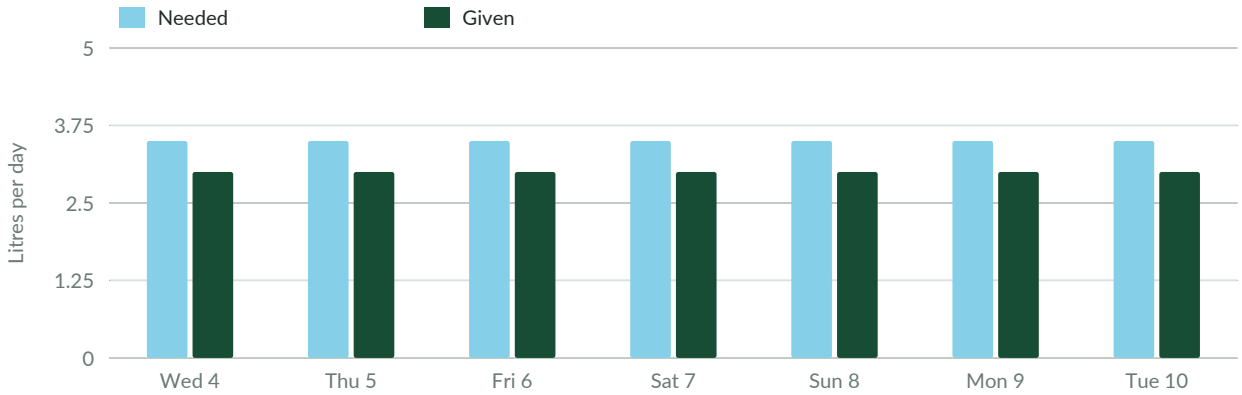
3 **Peak growth in Tub 1 for about 5 more weeks**: this is when shortages cost the most harvest, so don't let it dry out.
High confidence
Why: Tomato is at peak growth, when a shortage of water costs the most harvest. (note 3)

How your farm did this week

Zone	Water given	Water needed	Verdict	Pump runs	Rain
Tub 1 (Tomato)	about 21 L	about 24 L	Watered about right	7	0 mm

Why: The water given this period was close to what tomato needed. (note 4)

Did my crop get the water it needed?

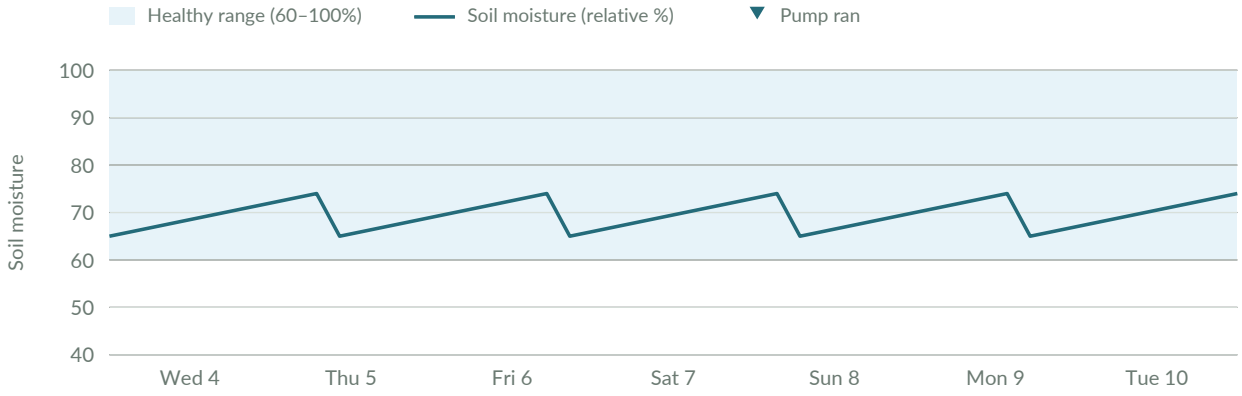


Yes - about 21 L against about 24 L needed this week.

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Was my soil in the healthy range?



Yes - it stayed in the healthy range every day this week.

What's coming this week?

Wed 11 Feb	Thu 12 Feb	Fri 13 Feb	Sat 14 Feb	Sun 15 Feb	Mon 16 Feb	Tue 17 Feb
31 °C	32 °C	33 °C	34 °C	35 °C	36 °C	37 °C
Rain 0 mm	Rain 0 mm	Rain 8 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm
Water Tub 1	No watering needed	Rain covers it	No watering needed	No watering needed	No watering needed	No watering needed

Area weather vs at your plants

Day	Area high	At the plants	Area humidity	At the plants
Wed 4 Feb	27 °C	29 °C	55%	60%
Thu 5 Feb	27 °C	29 °C	55%	60%
Fri 6 Feb	27 °C	29 °C	55%	60%
Sat 7 Feb	27 °C	29 °C	55%	60%
Sun 8 Feb	27 °C	29 °C	55%	60%
Mon 9 Feb	27 °C	29 °C	55%	60%
Tue 10 Feb	27 °C	29 °C	55%	60%

Only the days your tub sensor reported are shown.

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Technical appendix

For extension officers and markers. Figures and sources behind the advice; FAO-56 is Allen et al. (1998), page numbers as printed.

Tub 1

Figure	Value	Source
Relative moisture	17%	soil sensor reading 2700 against this zone's wet/dry setup - relative only, not a volume of water
Soil type	Loam / garden soil	FAO-56 Table 19 (Loam): Table 19 'Loam': theta_FC 0.20-0.30, theta_WP 0.07-0.17. Midpoints used.
Root depth	0.30 m	limited by this zone's depth; FAO-56 Table 22 gives 0.7 m for tomato
Water the soil can hold	39 mm	FAO-56 Eq. 82 from Table 19 soil values and Table 22 root depth
Crop water need factor today	1.15	FAO-56 Eq. 66 across Table 11 stage lengths, Kc values from Table 12 (Kc start: Table 12 group row 'b. Vegetables - Solanum Family')
Reference water use today	5.2 mm	FAO-56 Eq. 6 (Penman-Monteith), inputs from weather_app
This crop's water use today	5.9 mm	FAO-56 Eq. 58: crop factor x reference water use
Dryness it can stand before watering	14 mm	FAO-56 Eq. 83 with the depletion fraction from Table 22, adjusted for today's water use and for soil type
How dry the soil is now	39 mm short of full	FAO-56 Eq. 85 water balance, started from the soil sensor reading
TAW	39 mm	FAO-56 Eq. 82, p. 162; Table 19 p. 144; Table 22 pp. 163-165
RAW	14.1 mm	FAO-56 Eq. 83, p. 162
p (adjusted)	0.4	FAO-56 Table 22 p. 163-165; p. 162 ETc adjustment; p. 167 soil adjustment
Kc	1.2	FAO-56 Table 12 pp. 110-114; Eq. 66 p. 132
Confidence	High: your sensor, setup and the weather all agree.	ConfidenceRules (DATA RULE)

- FAO-56 publishes tomato Kc_end as the range 0.70-0.90 rather than a single value. The midpoint 0.80 is used, so late-season figures carry that spread.

- Stage lengths come from FAO-56 Table 11 (Table 11, Apr/May, Mediterranean), which the book says are indicative only - a real season may run shorter or longer.

- How dry the soil is was worked out from the sensor reading in proportion to what the soil can hold. The sensor measures relative wetness, not an actual volume of water, so this is an approximation.

- Water amounts in litres are estimates worked out from pump running time and the zone's flow rate. There is no flow meter anywhere in AgroFlow.

Assumptions

- Litres are estimated as pump running time x the zone's flow rate. There is no flow meter on any AgroFlow kit.

- Reference evapotranspiration (ETo) is FAO-56 Penman-Monteith (Eq. 6, p. 24) from the weather app. Solar radiation is always estimated from the day's temperature range (Eq. 50, p. 60); missing humidity is estimated from the temperatures (Eq. 64, p. 124); missing temperatures come from the tub's own sensor.

- Crop water need is ETc = Kc x ETo (Eq. 58, p. 103), with Kc from Table 12 joined by Eq. 66 (p. 132).

- The daily soil water balance is FAO-56 Eq. 85 (p. 170); stress days are days with Ks < 1 (Eq. 84, p. 169).

- Watering is advised when the root-zone depletion reaches RAW (p. 171): Dr >= RAW.

- Soil sensor readings are relative (wet/dry setup), not a measured volume of water.

- Verdicts: more than 1.2 x the need is too much, less than 0.8 x is too little (VerdictRules, DATA RULE).

Why we said this

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1. Your soil has dried to the point where tomato starts to struggle, and no rain is expected today.

- Crop: Tomato (this zone's setup)
- Crop stage: peak growth, day 81 of about 145 (your planting date)
- Soil: Loam / garden soil (this zone's setup)
- How dry the soil is: about 38.6 mm short of full (your soil sensor and weather app)
- Watering point: about 14.1 mm short of full (FAO crop water guide)
- Soil moisture: 17% (your soil sensor)
- Watering level: 2000 (this zone's setup)
- Rain expected today: 0 mm (weather app)
- Water your crop will use today: 5.9 mm (weather app and FAO crop water guide)
- How much: about 39 min (about 58 L) (this zone's setup)

The rule: We suggest watering once your crop has used about 36% of the water your soil can hold for it.
How sure: High: your sensor, setup and the weather all agree.
What would change it: If 5.9 mm of rain or more falls today, you can skip this.
Technical: Dr 38.6 mm, RAW 14.1 mm, TAW 39 mm, p 0.36, Kc 1.15, ETo 5.2 mm, ETc 5.9 mm. FAO-56 Eq. 58 (p. 103), 82-85 (pp. 162-170).
WaterRules.ShouldWaterToday (FAO-56 p. 171): Dr >= RAW.

2. The watering level saved for this zone waits until the soil is drier than tomato handles well.

- Watering level now: 2000 (this zone's setup)
- Recommended level: 1920 (FAO crop water guide)
- Crop: Tomato (this zone's setup)
- Crop stage: peak growth, day 81 of about 145 (your planting date)
- Soil: Loam / garden soil (this zone's setup)

The rule: Tomato can use about 40% of the water in the soil before it struggles. Your level waits longer than that.
How sure: High: your sensor, setup and the weather all agree.
What would change it: Tap Reset to recommended, or set the level to 1920 or wetter.
Technical: WaterRules.RecommendedThreshold: Table 22 p x soil factor (p. 167), clamped 0.1-0.8 (p. 162); recommended 1920, current 2000.

3. Tomato is at peak growth, when a shortage of water costs the most harvest.

- Crop: Tomato (this zone's setup)
- Crop stage: peak growth, day 81 of about 145 (your planting date)
- Soil: Loam / garden soil (this zone's setup)

The rule: Running short of water while a crop flowers and forms its harvest costs more than at any other time.
How sure: High: your sensor, setup and the weather all agree.
What would change it: This message stops when the crop moves into its last stage.
Technical: CropRules.SensitiveStage: FAO-56 p. 176 (flowering and yield formation); mid-season p. 95-96.

4. The water given this period was close to what tomato needed.

- Water given: about 21 L (pump run times)
- Water needed: about 24 L (weather app and FAO crop water guide)
- Verdict: About right (the rule below)

The rule: More than 1.2 times the need is too much and less than 0.8 times is too little. Pump amounts are estimates from run time and flow rate, so smaller differences are not flagged.
How sure: Medium: water given is estimated from pump run time and the zone's flow rate - there is no flow meter.
What would change it: A big change in the weather or in how long the pump runs.
Technical: VerdictRules.For(21 L, 24.5 L) = AboutRight; need = sum of ETc x area (FAO-56 Eq. 58, p. 103).